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| United States Patent | 12397201           |
| Kind Code            | B2                 |
| Date of Patent       | August 26, 2025    |
| Inventor(s)          | Lo; Chia Yu et al. |

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### Resistance adjustable power generator

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#### Abstract

a resistance adjustable power generator having: an operating unit that receives commands of a user and is able to switch the resistance adjustable power generator to either a release mode or a resistance mode; a resistance generating unit that receive the commands from the operating unit and provides adjustable multiple levels of resistance to a connecting device, with the resistance generating unit having a resistance controlling center, multiple resistance generator, and an output controlling center; and a battery electrically connects to the resistance generating unit.

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| <b>Family ID:</b> | <b>1000008780020</b>                                               |
| <b>Appl. No.:</b> | <b>18/456901</b>                                                   |
| <b>Filed:</b>     | <b>August 28, 2023</b>                                             |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20240075345 A1          | Mar. 07, 2024           |

#### Foreign Application Priority Data

|    |           |               |
|----|-----------|---------------|
| TW | 111133876 | Sep. 06, 2022 |
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#### Publication Classification

**Int. Cl.:** A63B24/00 (20060101); A63B21/00 (20060101); A63B21/005 (20060101); A63B22/00 (20060101); H02K7/108 (20060101); H02K7/116 (20060101); H02K7/18 (20060101); A63B71/06 (20060101)

## U.S. Cl.:

CPC     **A63B24/0087** (20130101); **A63B21/0058** (20130101); **A63B21/151** (20130101);  
**A63B22/0076** (20130101); **H02K7/108** (20130101); **H02K7/116** (20130101);  
**H02K7/1853** (20130101); A63B2071/0675 (20130101)

## Field of Classification Search

**CPC:**     A63B (24/0087); A63B (2024/0093); A63B (2024/0081); A63B (21/0058); A63B  
(21/151); A63B (21/0051); A63B (21/0052); A63B (21/0054); A63B (21/0055); A63B  
(21/225); A63B (21/0053); A63B (21/00181); A63B (21/0056); A63B (21/0057); A63B  
(21/00192); A63B (22/0076); A63B (22/0605); A63B (71/0619); A63B (2071/0675);  
A63B (2071/0081); A63B (2220/12); A63B (2220/62); A63B (2225/50); A63B  
(2069/167); A63B (69/16); H02K (7/108); H02K (7/116); H02K (7/1853); H02K  
(7/1861)

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## Background/Summary

### FIELD OF INVENTION

(1) The present invention relates generally to a power generator, and particularly to a resistance adjustable power generator.

### BACKGROUND OF THE INVENTION

(2) In today's health-conscious society, regular exercise has become an indispensable part of people's lives to improve physical fitness and overall well-being. However, the effectiveness of traditional fitness devices is hindered by a significant challenge of their unwieldy and bulky nature. This often discourages individuals from incorporating exercise into their daily routines due to space constraints and lack of convenience. Moreover, a considerable amount of energy is expended during workouts, only to be wasted without any means of harnessing or storing it. To address these pressing issues, there is a growing demand for the development of a compact, portable, and energy storing fitness device that is capable of efficiently storing the energy generated during exercise, making a positive contribution to environmental sustainability.

### SUMMARY OF THE INVENTION

(3) Therefore, it is a primary objective of the present invention to provide a resistance adjustable

power generator comprising an operating unit, a resistance generating unit, and a battery, wherein: the operating unit receives commands of a user and exchanges signals or the commands with the resistance generating unit, and the operating unit is able to switch the resistance adjustable power generator to either a release mode or a resistance mode, wherein in the resistance mode, the resistance generating unit provides multiple levels of resistance to a connecting device; and the resistance generating unit receive the commands from the operating unit and provides adjustable multiple levels of resistance to a connecting device, wherein the resistance generating unit comprises a resistance controlling center, multiple resistance generator, and an output controlling center, wherein the resistance controlling center controls movements of a clutch, controls the status of multiple resistance generators, and controls an electrical energy output by an output controlling center to tune multiple levels of the resistance, wherein: the clutch engages with the connecting device in the resistance mode and creates a first level of the resistance, and the clutch disengages from the connecting device in the riding mode, wherein the clutch is connected to an accelerator, and the accelerator increases the rotational speed of a coil winding in the resistance generator; the resistance generator connects to the clutch and comprises the coil winding, wherein the coil winding generates electrical energy during rotation; and the output controlling center regulates output values of voltage and/or current to modify the load output and generates adjustable multiple levels of the resistance and stores the electrical energy into the battery.

(4) Wherein, the clutch connects to the accelerator, and the accelerator comprises a hub gear mounted on a hub of the connecting device and a resistance gear mounted on the resistance generator, wherein the hub gear and the resistance gear are arranged to engage with each other directly or indirectly, and the number of teeth of the hub gear is greater than that of the resistance gear.

(5) Wherein, the accelerator comprises at least one planetary gearset.

(6) Wherein, the connecting device is a rear wheel of a bicycle, a pulley, or a flywheel.

(7) Wherein, the operating unit is a human-machine interface in a form of buttons, controllers, touch display panels, or mobile devices, and the operating unit receives inputs from a user, generates multiple commands, and sends the commands to the control center wirelessly or wired to tune multiple levels of the resistance.

(8) Wherein, the resistance generating unit comprises a resistance device; and the output controlling center adjust the resistance of the resistance device based on the electrical energy produced by the resistance generator.

(9) Wherein, the operating unit further comprises a course database with multiple exercise courses, wherein the operating unit adjusts settings of the resistance generating unit based on predefined parameters of simulation conditions associated with the exercise course or parameters specified by an instructor of the exercise course.

(10) Wherein, the resistance adjustable power generator is used in a fitness device.

(11) Wherein, the fitness device is a rowing machine or a chest press machine.

(12) The resistance adjustable power generator of this invention can be applied to various fitness devices to provide resistance training of the user, and thus greatly reduces the sizes of the fitness devices and provides consistent resistance and harvests electricity at the same time during the training of the user.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of a preferred embodiment of a resistance adjustable power generator in accordance with this invention;

(2) FIG. 2 is a block diagram of a preferred embodiment of a resistance adjustable power generator

in accordance with this invention;

(3) FIG. 3 is a block diagram of a preferred embodiment of a resistance adjustable power generator in accordance with this invention;

(4) FIG. 4 is a block diagram of a preferred embodiment of a resistance adjustable power generator in accordance with this invention;

(5) FIG. 5 is a schematic diagram of a preferred embodiment of a resistance adjustable power generator in accordance with this invention;

(6) FIG. 6 is a schematic diagram of a preferred embodiment of a resistance adjustable power generator in accordance with this invention;

(7) FIG. 7 is a schematic diagram of a preferred embodiment of a rowing machine with a resistance adjustable power generator in accordance with this invention; and

(8) FIG. 8 is a schematic diagram of a preferred embodiment of a chest press machine with a resistance adjustable power generator in accordance with this invention.

#### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) In order to make purposes, technical solutions, and advantages of the present invention to be clearer, the following content provides some preferred embodiments in accordance with the present invention.

(10) With reference to FIG. 1 to FIG. 4, a resistance adjustable power generator in accordance of this invention comprises an operating unit **30**, a resistance generating unit **10**, and a battery **20**. The resistance generating unit **10** provides adjustable resistance to a connecting device (not shown). The resistance generating unit **10** is electrically connected to the battery **20**, and the resistance generating unit **10** and the operating unit **30** can exchange messages mutually.

(11) The term “electrically connected” can be either a wired or wireless connection. The term “mutual message exchange” refers to the exchange of data or commands between the operating unit **30** and the resistance generating unit **10**.

(12) The operating unit **30** receives the commands in the form of a user operation. The operating unit **30** can take various forms, such as a button, a controller, a touch display panel, a mobile device, or any other human-machine interface (HMI) that can display information and send the commands, and the operating unit **30** can input commands or display output results.

(13) The operating unit **30** is used to receive signals or commands from the resistance generating unit **10** and the battery **20**, and the operating unit **30** also generates corresponding responses and output controls accordingly. For example, the operating unit **30** can drive the resistance generating unit **10** to provide resistance and charge the battery **20**. The signal received by the operating unit **30** can include information such as the remaining power and health status of the battery **20** or abnormal messages generated by the resistance generating unit **10**. The operating unit **30** can then provide feedback to the user based on the received signals.

(14) The operating unit **30** can further connect to a terminal device of the user via wireless network transmission technology and synchronously transmit the signals or the commands to the terminal device.

(15) Preferably, the operating unit **30** can utilize GNSS signals, such as GPS signals, and through network transmission to achieve real-time monitoring. For example, the operating unit **30** can be a controller with a universal communication protocol, such as a controller that follows the Controller Area Network (CANBUS), Bluetooth Low Energy (BLE), ANT+, or Fitness Machine Service (FTMS) protocols. The operating unit **30** communicates with the battery **20** and the resistance generating unit **10** through the protocol(s), either wirelessly or wired.

(16) The resistance generating unit **10** reads the signals from the operating unit **30**, obtains the power status signal from the battery **20**, and adjusts output of resistance based on the multiple signals received. Furthermore, the resistance generating unit **10** can be connected to the terminal device of the user wirelessly, and synchronously transmit the signals or commands to the terminal device.

(17) The resistance adjustable power generator in accordance with this invention includes a release mode and a resistance mode. The user can give instructions to the operating unit **30** or the terminal device such as a mobile device to switch between the release mode and resistance mode of the resistance adjustable power generator according to the user.

(18) When the resistance adjustable power generator is in the release mode, the user can use it without activating the resistance generating unit **10**. The battery **20** can still output electric energy in the release mode.

(19) When the resistance adjustable power generator is in the resistance mode, the operating unit **30** activates the resistance generating unit **10** to establish an electrical connection with the battery **20**. As the resistance generating unit **10** is activated, the resistance generating unit **10** converts mechanical energy generated from the user that transmits through the connecting device into electrical energy, and the electrical energy is then stored in the battery **20** as chemical energy. This allows the user **10** to charge the battery **20** while working out, making the resistance adjustable power generator of this invention an energy-efficient and environmentally-friendly option.

(20) Moreover, the resistance adjustable power generator can automatically switch between the release mode and the resistance mode. For example, when the user starts to work out, the resistance adjustable power generator can automatically switch to the resistance mode and activates the resistance generating unit **10**, making the resistance generating unit **10** instantly convert energy from the user and charge the battery **20**. After the user stops workout for a while, the resistance adjustable power generator can automatically switch to the release mode and turn off the resistance generating unit **10**.

(21) Similarly, the user can still decide whether the resistance adjustable power generator should automatically switch between the release mode and the resistance mode with the operating unit **30**. This can prevent the resistance adjustable power generator from repeatedly switching between the release mode and the resistance mode.

(22) The resistance generating unit **10** is in the form of a generator. The generator can be a magnetic reluctance device, a contact friction device, or a combination of both. The resistance generating unit **10** provides rotational resistance to the connecting device, and thus configured the connecting device with multiple levels of resistance in the resistance mode to meet various resistive force requirements.

(23) The connecting device can be a rear wheel of a bicycle, a pulley, or a flywheel. In a preferred embodiment, the connecting device is a rear wheel of an electric bicycle, and the resistance generating unit **10** is placed at the rear wheel of the electric bicycle. In this embodiment, the resistance generating unit **10** adjusts the output load of the multi-level resistance by regulating the voltage and/or current output, or by turning on/off multiple coil windings according to meet the demand for different scales of resistance. This enables the adjustment of multiple level outputs of resistance. The magnetic reluctance device of the resistance generating unit **10** can generate magnetic resistance on the wheel of the electric bicycle by utilizing the interaction between magnets and eddy currents, and the resistance generating unit **10** can regulate the output voltage and/or current, or turn on/off multiple coil windings to adjust the generated magnetic resistance. The contact friction device of the resistance generating unit **10** directly provides different levels of rotational resistance to the connecting device, such as the rear wheel of the bicycle, the pulley, or the flywheel, by direct contact.

(24) With reference to FIG. **1** to FIG. **4**, the resistance generating unit **10** comprises a resistance controlling center **11**, a resistance generator **12**, and an output controlling center **13** that are electrically interconnected. The resistance controlling center **11** receives signals transmitted from the operating unit **30**, determines whether the resistance generator **12** should be activated, and determines the amount of electrical energy to be outputted by the output controlling center **13** to the battery **20**.

(25) The resistance generator **12** in the present embodiment includes a coil winding. The resistance

generator **12** produces the electrical energy by the rotation of a hub of the connecting device relative to the coil winding and simultaneously generates rotational resistance on the connecting device. The resistance generator **12** is able to provide the connecting device multiple levels of resistance in the resistance mode.

(26) In a preferred embodiment in accordance with this invention as shown in FIG. 2, the resistance generating unit **10** includes several resistance generators **12**, and the number of the resistance generators **12** to be turned on or turned off can be determined based on the level of resistance set in the resistance mode. The more resistance generators **12** that are turned on, the greater the rotational resistance and the greater the resistance applied to the connecting device, resulting in greater resistance.

(27) The output controlling center **13** is further used to regulate and set the output value of voltage and/or current. The output controlling center **13** can allocate the electrical energy generated by the resistance generator **12** and the energy output from the battery **20**. The output controlling center **13** can also modify the output of electric load (i.e., output load) through voltage transformation and/or rectification, thereby adjusting the resistance levels during the rotation of the hub to achieve different levels of resistance.

(28) By tuning the on and off of multiple resistance generators **12** or adjusting the output of the output controlling center **13**, the resistance adjustable power generator can possess variable resistance modes with multiple resistance levels. This allows the electric bicycle **10** to offer various levels of resistance for different workout experiences.

(29) In a preferred embodiment, as shown in FIG. 3, the resistance generating unit **10** is positioned on the hub of the connecting device, and the resistance generating unit **10** is detachably connected to the hub. Moreover, the resistance generating unit **10** is equipped with a clutch **14**. The clutch **14** operates according to the mode specified by the resistance controlling center **11**. In the release mode, the clutch **14** disengages from the connecting device and disconnects the resistance generators **12** from the connecting device, reducing the resistance of the connecting device. When the resistance adjustable power generator switches to the resistance mode, the clutch **14** engages with the connecting device and creates a first level of riding resistance.

(30) In a preferred embodiment shown in FIG. 3 and FIG. 5, the resistance generating unit **10** includes an accelerator **15**. The accelerator **15** is designed to increase the rotational speed of the coil winding in the resistance generator **12**. Preferably, the accelerator **15** includes a hub gear **151** mounted on the hub and a resistance gear **152** mounted on the resistance generator **12**, wherein the hub gear **151** and the resistance gear **152** are arranged to engage with each other directly or indirectly. By adjusting the gear ratio between the hub gear **151** and the resistance gear **152** (e.g., by having a much greater number of teeth on the hub gear **151** compared to the resistance gear **152**), the overall rotational speed of the resistance generator **12** can be significantly greater than that of the hub gear **151**. In this preferred embodiment, an idler gear **154** is provided between the hub gear and the resistance generator **12** to prevent any interference between the components. This idler gear **154** ensures smooth and efficient transmission of rotational motion between the hub gear **151** and the resistance generator **12**.

(31) With reference to FIG. 6, a preferred embodiment of the resistance generating unit **10** is installed on the hub of the connecting device and the resistance generating unit **10** also includes the resistance generator **12**, the clutch **14**, and the accelerator **15**. The accelerator **15** is positioned between the resistance generator **12** and the clutch **14**. Preferably, the accelerator **15** includes at least one planetary gearset **153**. In a preferred configuration, the accelerator **15** consists of two parallel planetary gearsets **1553** positioned next to the resistance generator **12**. In the resistance mode, the clutch **14** engages with the hub of the connecting device. The hub is coupled with a chain or a cable of the connecting device and the hub is driven by the chain or the cable of the connecting device, and the planetary gearset **153** rotates in accordance with the rotation of the hub of the connecting device; thereby the planetary gearset **1553** drives the operation of the resistance

generator **12** and greatly increase the rotation speed of the coil winding of the resistance generator **12**. This configuration of the planetary gearset **153** enables a significant reduction in overall size of the resistance generator **12** while achieving increased efficiency in generating resistance and generating electricity. In the release mode, the clutch **14** disengages from the hub of the connecting device and reduces rotational resistance of the connecting device.

(32) With reference to FIG. **4**, in a preferred embodiment, the resistance generating unit **10** can also include a resistance device **16**, which directly generates resistance to avoid insufficient resistance provided by the resistance generator(s) **12**, ensuring sufficient training functionality. The resistance controlling center **11** receives signals from the operating unit **30** and activates the resistance generator **12** based on the settings of the resistance mode. In this preferred embodiment, the output controlling center **13** can determine the activation and tuning of the resistance device **16** based on the resistance generated by the resistance generator **12**. This ensures that the resistance generating unit **10** can provide stable resistance in the resistance mode, regardless of the force or speed the user pulls the cable or the chain or the amount of energy generated by the resistance generator **12**.

(33) Furthermore, in another preferred embodiment, the resistance generating unit **10** serves as a braking assistance device. For example, during braking of the bicycle, the resistance generating unit **10** receives a brake signal from the bicycle. The resistance controlling center **11** determines the level of resistance to be generated by either the resistance generating unit **10** or the output controlling center **13** based on the brake signal. In another example, the resistance controlling center **11** determines the level of resistance to be generated by either the resistance generating unit **10** or the output controlling center **13** based on the workout conditions of the user.

(34) Preferably, the operating unit **30** also includes a course database, which contains several exercise courses generated based on simulation conditions. The user can select an exercise course through the operating unit **30**. The operating unit **30** adjusts the settings of the resistance generating unit **10** based on predefined parameters of the simulation conditions associated with the exercise course or parameters specified by an instructor of the exercise course. This adjustment can be done by modifying the configuration parameters of the resistance generating unit **10**.

(35) Furthermore, the user can also adjust the parameters of the exercise course according to preferences using the operating unit **30**. This allows for customization of the exercise course, tailoring the exercise course to the user's specific needs and goals.

(36) The operating unit **30** also includes a data storage unit. The data storage unit is used to store the execution records and parameters of each exercise course. The execution records may include the duration of the exercise course, rest periods during the process, calories burned, the number of executions, and historical data.

(37) Furthermore, the data storage unit can also store the records from the resistance generating unit **10**.

(38) Furthermore, the resistance generating unit **10** and the operating unit **30** can store data in a cloud server through internet connection. This allows the user to connect through the terminal device and access the data directly.

(39) Furthermore, when the exercise course is executed, the operating unit **30** can also choose to activate the resistance mode and initiate the resistance generating unit **10**. Therefore, the resistance adjustable power generator of this invention can simultaneously charge the battery **20** through the resistance generating unit **10** while the exercise course is executed.

(40) Furthermore, the operating unit **30** can retrieve the record, display the record, and assist the user in achieving daily training goals.

(41) With reference to FIG. **7**, a preferred embodiment of a rowing machine **40** with the resistance adjustable power generator in accordance with this invention is shown. In this embodiment, the rowing machine has the operating unit **30**, the battery **20** (not shown), the resistance generating unit **10**, the connecting device coupled with the resistance generating unit **10**, a chain (not shown) coupled with the connecting device, and a handle **41** connected to the chain. In this embodiment,

the user can sit on a seat **43** and step on footrests **44** of the rowing machine **40**, and the seat **43** can linearly move along a slide **42** of the rowing machine **40**.

(42) In this preferred embodiment, the operating unit **30** is in the form of a smart device with a touch panel, and the connecting device is the flywheel connected to the resistance generating unit **10**. When the user starts to workout, the operating unit **30** sends the commands received from the user and activates the resistance mode of the resistance adjustable power generator and adjusts the scale of the resistance according to the commands. When the resistance mode is activated, the clutch **14** of the resistance generating unit **10** engages with the flywheel of the rowing machine **40** and provides a first level of resistance. The user can further adjust the strength of the resistance, with the resistance controlling center **11** of the resistance generating unit **10** further activates the resistance generators **12** and increases the strength of the resistance. Moreover, the resistance generating unit **10** can further increase or decrease the resistance by regulating the voltage and/or current output of the resistance generators **12** with output controlling center **13** to provide a wide range of resistance and smooth adjustment (i.e., the difference between the scales of neighboring resistance is miniscule).

(43) In this preferred embodiment, the resistance of the resistance generating unit **10** is transmitted through the resistance generator **12** to the flywheel of the rowing machine **40**, and then the resistance is further transmitted to the chain coupled with the flywheel and finally reaches the handles **41** attached to the chain. The electricity generated by the resistance generating unit **10** is then stored in the battery **20**, and the battery **20** can provide electricity to the operating unit **30** of the rowing machine and may provide electricity to other devices of the user.

(44) With reference to FIG. **8**, a preferred embodiment of a chest press machine **50** with the resistance adjustable power generator in accordance with this invention is shown. In this embodiment, the resistance adjustable power generator of the chest press machine **50** has the operating unit **30** (not shown), the battery **20** (not shown), the resistance generating unit **10**, the connecting device coupled with the resistance generating unit **10**, a cable **51** coupled with the connecting device, multiple pulleys **52**, and a bar handle **53** connected to the cable **51**.

(45) In this preferred embodiment, the operating unit **30** is in the form of a smartphone of the user, and the connecting device is the flywheel connected to the resistance generating unit **10**. When the user starts to workout, the operating unit **30** sends the commands received from the user and activates the resistance mode of the resistance adjustable power generator and adjusts the scale of the resistance according to the commands. When the resistance mode is activated, the clutch **14** of the resistance generating unit **10** engages with the flywheel of the chest press machine **50** and provides a first level of resistance. The user can further adjust the strength of the resistance, with the resistance controlling center **11** of the resistance generating unit **10** further activates the resistance generators **12** and increases the strength of the resistance. Moreover, the resistance generating unit **10** can further increase or decrease the resistance by regulating the voltage and/or current output of the resistance generators **12** with output controlling center **13** to provide a wide range of resistance and smooth adjustment (i.e., the difference between the scales of neighboring resistance is miniscule).

(46) In this preferred embodiment, the resistance of the resistance generating unit **10** is transmitted through the resistance generator **12** to the flywheel of the chest press machine **50**, and then the resistance is further transmitted to the cable **51** coupled with the flywheel and passing through multiple pulleys **52** to alter the direction of the resistive force, and finally reaches the bar handles **53** attached to the cable **51**. The electricity generated by the resistance generating unit **10** is then stored in the battery **20**, and the battery **20** can provide electricity to other devices of the user.

(47) The resistance adjustable power generator in accordance with this invention can be applied to various fitness devices to provide resistance training of the user. The resistance adjustable power generator in accordance with this invention greatly reduces the sizes of the fitness devices and



provides consistent resistance and harvests electricity at the same time during the training of the user.

## Claims

1. A resistance adjustable power generator comprising an operating unit, a resistance generating unit, and a battery, wherein: the operating unit receives commands of a user and exchanges signals or the commands with the resistance generating unit, and the operating unit is able to switch the resistance adjustable power generator to either a release mode or a resistance mode, wherein in the resistance mode, the resistance generating unit provides multiple levels of resistance to a connecting device; and the resistance generating unit receive the commands from the operating unit and provides adjustable multiple levels of resistance to the connecting device wherein the resistance generating unit comprises a resistance controlling center, multiple resistance generator, and an output controlling center, wherein the resistance controlling center controls movements of a clutch, controls the status of the multiple resistance generator, and controls an electrical energy output by the output controlling center to tune multiple levels of the resistance, wherein: the clutch engages with the connecting device in the resistance mode and creates a first level of the resistance, and the clutch disengages from the connecting device in the riding mode, wherein the clutch is connected to an accelerator, and the accelerator increases the rotational speed of a coil winding in the resistance generator; the resistance generator connects to the clutch and comprises the coil winding, wherein the coil winding generates electrical energy during rotation; and the output controlling center regulates output values of voltage and/or current to modify the load output and generates adjustable multiple levels of the resistance and stores the electrical energy into the battery.
2. The resistance adjustable power generator according to claim 1, wherein the clutch connects to the accelerator, and the accelerator comprises a hub gear mounted on a hub of the connecting device and a resistance gear mounted on the resistance generator, wherein the hub gear and the resistance gear are arranged to engage with each other directly or indirectly, and the number of teeth of the hub gear is greater than that of the resistance gear.
3. The resistance adjustable power generator according to claim 2, wherein the operating unit is a human-machine interface in a form of buttons, controllers, touch display panels, or mobile devices, and the operating unit receives inputs from a user, generates multiple commands, and sends the commands to the control center wirelessly or wired to tune multiple levels of the resistance.
4. The resistance adjustable power generator according to claim 2, wherein the operating unit further comprises a course database with multiple exercise courses, wherein the operating unit adjusts settings of the resistance generating unit based on predefined parameters of simulation conditions associated with the exercise course or parameters specified by an instructor of the exercise course.
5. The resistance adjustable power generator according to claim 1, wherein the accelerator comprises at least one planetary gearset.
6. The resistance adjustable power generator according to claim 5, wherein the operating unit is a human-machine interface in a form of buttons, controllers, touch display panels, or mobile devices, and the operating unit receives inputs from a user, generates multiple commands, and sends the commands to the control center wirelessly or wired to tune multiple levels of the resistance.
7. The resistance adjustable power generator according to claim 6, wherein: the resistance generating unit comprises a resistance device; and the output controlling center adjust the resistance of the resistance device based on the electrical energy produced by the resistance generator.
8. The resistance adjustable power generator according to claim 6, wherein the operating unit further comprises a course database with multiple exercise courses, wherein the operating unit

adjusts settings of the resistance generating unit based on predefined parameters of simulation conditions associated with the exercise course or parameters specified by an instructor of the exercise course.

9. The resistance adjustable power generator according to claim 6, wherein the resistance adjustable power generator is used in a fitness device.

10. The resistance adjustable power generator according to claim 9, wherein the fitness device is a rowing machine with a chain coupled to the connecting device.

11. The resistance adjustable power generator according to claim 9, wherein the fitness device is a chest press machine with a cable coupled to the connecting device.

12. The resistance adjustable power generator according to claim 5, wherein the operating unit further comprises a course database with multiple exercise courses, wherein the operating unit adjusts settings of the resistance generating unit based on predefined parameters of simulation conditions associated with the exercise course or parameters specified by an instructor of the exercise course.

13. The resistance adjustable power generator according to claim 5, wherein the resistance adjustable power generator is used in a fitness device.

14. The resistance adjustable power generator according to claim 1, wherein the connecting device is a rear wheel of a bicycle, a pulley, or a flywheel.

15. The resistance adjustable power generator according to claim 14, wherein the resistance adjustable power generator is used in a fitness device.

16. The resistance adjustable power generator according to claim 1, wherein the operating unit is a human-machine interface in a form of buttons, controllers, touch display panels, or mobile devices, and the operating unit receives inputs from a user, generates multiple commands, and sends the commands to the control center wirelessly or wired to tune multiple levels of the resistance.

17. The resistance adjustable power generator according to claim 1, wherein the operating unit further comprises a course database with multiple exercise courses, wherein the operating unit adjusts settings of the resistance generating unit based on predefined parameters of simulation conditions associated with the exercise course or parameters specified by an instructor of the exercise course.

18. The resistance adjustable power generator according to claim 1, wherein the resistance adjustable power generator is used in a fitness device.

19. The resistance adjustable power generator according to claim 18, wherein the fitness device is a rowing machine.

20. The resistance adjustable power generator according to claim 18, wherein the fitness device is a chest press machine.

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